

Week 6 — Complex Survey Analysis: Why NHANES Is Not a Simple Random Sample

SPPH 604 · Application of Advanced Epidemiological Methods

Running example: Kueh et al. (2026), MASLD / NHANES

Learning objectives

- ▶ Explain why **NHANES** is a stratified, multistage, clustered probability sample — and why that breaks default standard errors.
- ▶ Name the three design objects every analysis needs: **weights** (WTINT/WTMEC/WTSAF), **strata** (SDMVSTRA), **PSU** (SDMVPSU).
- ▶ Pick and **pool** the correct weight when combining cycles; state what **design effects** do to confidence intervals.
- ▶ Contrast the paper's **unweighted** estimates with **design-aware** ones — and see why unweighted was a *choice*.
- ▶ Reason about the **fasting-vs-non-fasting** weight puzzle that MASLD's own analyte mix creates.

Continuity: where Week 5 left us

- ▶ **Recap Wk5:** the targeted IV-vs-III $\times$ sex test gave a ratio of HRs of **1.00 (0.42–2.35)** — underpowered rather than resolved.
- ▶ Lesson: a crude stratum difference is **not an interaction test**, and a null test is **not evidence of homogeneity** — a null with its uncertainty attached, not a demonstration that sex is irrelevant (P3).
- ▶ **Today:** a different threat — not confounding, not modification, but whose *population* the estimates describe.
- ▶ The paper analyzed NHANES **as if it were a simple random sample**.
- ▶ **Sets up L5** (naive vs survey-weighted) and **P4**, your first design-aware re-estimate.

Where we are in the project

- ▶ Course arc: **Replicate** → **Interrogate** → **Improve** → **Defend**.
- ▶ **M1** replication due Wk8; **M2** critique of another group's M1 Wk9.
- ▶ Done so far — **P1** eligibility, **P2** Table 1, **P3** effect modification — all built on an *unweighted* analytic file.
- ▶ Today we question a **design assumption baked into every number so far**.

MASLD: the paper reports HRs to two decimals but does not name the survey design.

The motivating puzzle

- ▶ The paper's headline: **Group IV** all-cause **HR 15.1** (unadjusted).
- ▶ Ask the class: **15.1 for whom** — the *sample*, or *US adults* with MASLD?
- ▶ That gap between **sample** and **population** is exactly what survey design bridges.
- ▶ Punchline preview: design-aware weighting pulls Group IV toward **12.4** and **widens** the CI.

MASLD: the inference target is implicitly the US population, but the math is the sample's.

What “simple random sample” assumes

- ▶ **SRS:** every person drawn *independently* and with *equal* probability.
- ▶ Default SEs — including coxph's — assume exactly this.
- ▶ Two violations are coming:
 - ▶ **Unequal selection probabilities** → handled by *weights*.
 - ▶ **Non-independence** (clustering) → handled by *strata/PSU*.
- ▶ Ignore them and: point estimates tilt toward **sample composition**; SEs run **too small**.

MASLD: the default coxph SEs in the replication silently assume SRS.

How NHANES is actually built

- ▶ **Multistage** selection: (1) counties = **PSUs** → (2) segments → (3) households → (4) persons.
- ▶ **Stratified** first by design domain; deliberate **oversampling** of targeted subgroups (older adults, some race/ethnicity groups, lower income).
- ▶ Result: neither equal probabilities *nor* independent draws — **both** SRS assumptions fail *by design*.
- ▶ This is a feature, not a flaw — it buys precision where policy needs it.

MASLD: MASLD adults are not sprinkled randomly across PSUs — they cluster with the sampled households.

Weights: WTINT, WTMEC, WTSAF

- ▶ A weight **how many people in the target population** this respondent represents.
- ▶ Formally: **inverse selection probability**, adjusted for nonresponse and post-stratification.
- ▶ Three nested tiers by how deep into the exam a variable lives:
 - ▶ WTINT (interview) WTMEC (mobile exam) WTSAF (fasting subsample).
- ▶ **Rule:** use the weight for the **most restrictive** data source your analysis actually needs.

MASLD: FLI needs **triglycerides** — a fasting analyte — so the honest weight is WTSAF, the smallest, most-adjusted tier.

Strata and PSUs: SDMVSTRA, SDMVPSU

- ▶ SDMVSTRA encodes the **design strata**; SDMVPSU the (masked) **primary sampling units** within a stratum.
- ▶ These drive **variance, not the point estimate**.
- ▶ Clustering: neighbors resemble each other → **effective N** < **row count**.
- ▶ Analytic strata/PSU are **pseudo-values** released for disclosure control — use *as provided*, never re-derive.

MASLD: none of SDMVSTRA / SDMVPSU appear anywhere in the paper's methods.

Pooling weights across cycles

- ▶ NHANES 2007–2018 = **six** 2-year cycles; the paper stacks all six.
- ▶ Weights are built **per cycle** → divide by the number of cycles pooled (here /**6**) so the weighted population total isn't multiplied.
- ▶ Use the **cycle-appropriate** weight variable and pool consistently.
- ▶ **Strata/PSU** are pooled **as-is** — no division.

MASLD: with six cycles combined, the correct pooled fasting weight is **WTSAF/6** — a step easy to miss.

The design effect (DEFF)

- ▶ **DEFF** variance under the real design $\div$ variance under SRS.
- ▶ Typically > 1 because of clustering + unequal weights.
- ▶ Practical read: when **DEFF** > 1 , your **effective N is smaller** than your row count $\rightarrow$ honest CIs get **wider**.
- ▶ “**N = 6,300**” overstates the information content once design is respected.

MASLD: the paper's tight CIs partly reflect pretending each of ~6,300 rows is an independent draw.

Short break (5 min)

- ▶ Stretch, water, reset.
- ▶ Coming up:
 - ▶ what happens to the **MASLD numbers** when we do it right, and
 - ▶ the genuine **fasting-weight puzzle** MASLD creates.

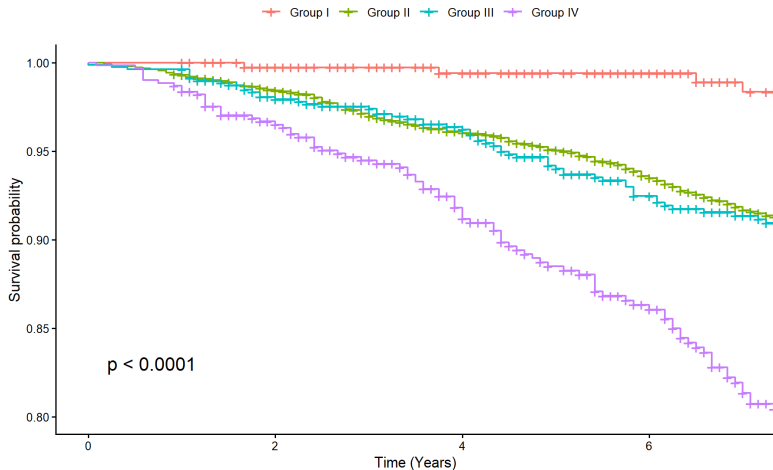
“Unweighted” is a choice, not a default

- ▶ Some analysts defend **unweighted, model-based** inference when the outcome model is richly adjusted (a *conditional*, not *descriptive*, target).
- ▶ But that stance must be **stated and defended** — silence reads as oversight, not decision.
- ▶ **Two separate axes, not two rival framings:**
 - ▶ **Design** decides *which population* the estimate speaks for.
 - ▶ **Adjustment** decides *which covariates* it holds fixed.
 - ▶ You **cross** them — crude/adjusted $\times$ unweighted/design-aware, on the **same rows**.
- ▶ **Declare the estimand** — population-average, conditional on covariates, or both — name the weight and **which cell of the 2 $\times$ 2** your headline came from.

MASLD: the paper never declares which target it's after — so readers can't tell if unweighted was *reasoned* or *accidental*.

What weighting does to MASLD

All-Cause Mortality



Number at risk

391	356	291	221
4585	4134	3272	2501
787	732	637	522

Two axes, four cells

All four cells on the **same** Model-1 complete-case rows (**n = 5,939**).

	IV vs I (<i>the paper's contrast</i>)	IV vs III (<i>our locked contrast</i>)
unweighted, crude	16.05 (5.91–43.58)	2.34 (1.72–3.18)
unweighted, Model 1	3.79 (1.38–10.39)	1.06 (0.77–1.45)
weighted, crude	11.93 (2.60–54.84)	3.07 (2.11–4.48)
weighted, Model 1	2.97 (0.69–12.79)	1.29 (0.90–1.86)

- ▶ Read **down** a column for **adjustment**; **across** a row for **design**.
- ▶ The familiar ledger — unadjusted, then fully adjusted, then survey-weighted — mixes crude and adjusted models fitted on **different row sets**. It is **not** a sequence of successive corrections, and we do not present it as one.

The question settled by CDC's least-common-denominator rule: fasting vs non-fasting analytes

- ▶ **FLI** combines **triglycerides + GGT + BMI + waist**.
- ▶ **TG** is a *fasting* analyte (WTSF domain); **GGT** is *not* — it lives in the broader MEC (WTMEC).
- ▶ The exposure definition **straddles two subsamples** — but this is *not* contested: the rule settles it.
- ▶ **Least-common-denominator:** use the weight of the **smallest applicable** subsample → WTSF, because FLI needs a fasting analyte.
- ▶ **WTMEC is not a defensible *primary* weight here** — it weights to a population in which the exposure cannot be constructed at all.
- ▶ But **run it anyway, labelled as a sensitivity analysis**.
Showing how far the estimate moves under the wrong weight is how you demonstrate that the rule mattered.
- ▶ The job is to **apply the rule, say so, and show what was at stake** — one prespecified primary, one labelled sensitivity run.

Subset the design, never the data

- ▶ **Cardinal rule:** declare the *full-survey* design **once**, then `subset()` **within it**.
- ▶ **Never** filter rows first and re-declare a design on the survivors.
- ▶ Dropping rows first discards the strata/PSU information non-selected units contribute to **variance** → wrong SEs.
- ▶ This is the **#1 practical error** students hit in L5.

MASLD: eligibility (**FLI 60**, 1 cardiometabolic criterion) is a *subset* — apply it to the design, keeping WTSAF/strata/PSU intact.

Design-aware workflow, end to end

Six steps, in order: {.incremental}

1. **Pick the weight tier** by the most-restrictive variable.
2. **Pool** — divide by number of cycles (/6).
3. **Declare the design** with weight + strata + PSU.
4. **Subset within** the design (never the rows).
5. **Fit** the survey-aware model.
6. **Report DEFF-widened CIs** — plus a one-line design statement in every table caption.

► **Sensitivity analysis is part of the deliverable**, not optional polish. The weight has a settled primary (WTSAF), so the sensitivity run is the WTMEC comparison — reported and labelled, never swapped in as the headline.

MASLD: this six-step recipe is precisely what **P4** asks you to run on your own paper.

Common pitfalls checklist

- ▶ Forgetting to **divide pooled weights** by the number of cycles.
- ▶ **Filtering rows before declaring the design** — the fatal one.
- ▶ Using WTMEC when a **fasting analyte** is in the model (or vice versa).
- ▶ Reading **unweighted vs weighted** divergence as “error” rather than as **two different estimands**.

Bridge to today's lab (L5) and P4

- ▶ **L5 (EpiMethods Complex Survey module):** re-fit the MASLD Group I–IV Cox model **two ways** on the fixed dataset:
 - ▶ a **naive** unweighted fit, then
 - ▶ a **design-aware** fit — declare WTSAF/6 + SDMVSTRA + SDMVPSU, and **subset the design** (not the rows) to the MASLD-eligible sample.
- ▶ Tabulate **naive vs survey** HRs/CIs side by side; quantify the attenuation (**15** → **12.4**) and CI widening.
- ▶ **P4 deliverable — your design-aware re-estimate:** run the same six-step workflow on *your* paper's headline estimate, **declare and justify** your chosen weight (name the fasting-vs-non-fasting tension if it applies), report the **naive-vs-weighted contrast**, and write the **one-line design statement** the paper omitted.
- ▶ **Time-to-event route** (M0 criterion 5, no survey weights): those six steps are not yours, and there is **no naive-vs-correct number**. Reproduce the paper's survival estimand — follow-up time and censoring intact — then **interrogate the one structural choice you named at M0:**